Miechilly Duong

miechillyduong@gmail.com | (770)-669-0064 | US Citizen | linkedin.com/in/miechillyduong

EDUCATION

Kennesaw State University, Kennesaw, GA *Expected Graduation: May 2029*

- B.S. Civil Engineering | Concentration in Geotechnical, Construction, and Geostructural GPA: 3.6

Relevant Coursework: Survey|Civil Graphics|Survey Engr Apps Math|Differantial Equations|Calculus 3

PROFESSIONAL EXPERIENCE

Kennesaw State University|Korea Expressway Corporation (KEC), Marietta, GA *January 2026 - May 2026*

Undergraduate Researcher - KEC sponsored

- Conducted research on a $60,000 project investigating the balance between hydraulic/acoustic performance and the structural lifespan of permeable highway asphalt under heavy traffic and severe weather.
- Performed LCA and LCCA on sustainable concrete mixes using ISO-compliant methodologies and assessed the material's economic performance over its entire lifespan.
- Applied data analysis and technical communication skills to present findings at the Undergraduate Research Symposium at Kennesaw State University.

Kennesaw State University|Georgia Department of Transportation (GDOT), Marietta, GA *Aug 2026 - Present*

Undergraduate Researcher- GDOT sponsored

- Conducted research on a $335,000 GDOT project, RP 25-11, investigating higher-fines backfill materials for mechanically stabilized earth (MSE) walls.
- Executed hydraulic conductivity testing of coarse- and fine-grained soils using constant-head permeameter methods in accordance with ASTM D2434.
- Collected, analyzed, and documented lab data to evaluate relationships between hydraulic gradient and soil discharge velocity.
- Performed Finite Element (FE) simulations to determine MSE wall deformation for various backfill materials.

PROJECTS

Smart Calorie Scale (Arduino and MATLAB) *April 2026*

- Developed a smart food scale integrating an Arduino, HX711 load cell, and MATLAB for real-time weight measurement, calibration, data visualization, and caloric estimation.
- Troubleshot sensor calibration, signal noise, wiring, and load-cell sensitivity to improve system reliability.

Automatic Irrigation System (Arduino) *December 2025*

- Built an Arduino-based automatic irrigation system using soil-moisture sensors, adjustable moisture thresholds, and relay-controlled water pumps.

Fireplace Fabrication (AutoCAD) *April 2025*

- Fabricated a fireplace using AutoCAD, laser cutting, and E7018/E6010 welding with GD&T.

PROFESSIONAL INVOLVEMENT

- **ASCE (American Society of Civil Engineers)**: Member
- **Steel Bridge Competition Team**: Fabrication and welding

SKILLS AND SOFTWARE

- **Software**: AutoCAD, Civil 3D, MATLAB, Microsoft Excel, Google Workspace.
- **Languages**: English and Vietnamese